

Towards Integrating Human-in-the-loop Control in Proactive Intelligent Personalised Agents

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ABSTRACT

This research explores the integration of Human-in-the-loop (HITL) control within Proactive Intelligent Personalised Agents (PIPAs) that possess the capability to proactively anticipate users' needs and perform tasks on their behalf. The proactive assistance offered by PIPAs is tailored to individual users' preferences and behaviours. However, it is crucial to personalise the level of proactivity exhibited by PIPAs to align with users' preferences regarding the potential delegation of autonomy. This necessitates HITL control to regulate PIPAs' autonomy levels, ensuring appropriate user involvement in decision-making. Through a simulation-based approach, this research explores the integration of HITL control in PIPAs. It investigates the conditions that trigger HITL control, its mechanisms, and the challenges associated with these triggers.

CCS CONCEPTS

• **Human-centered computing** → *User models*.

KEYWORDS

Intelligent Personal Assistants, Human-in-the-loop, Proactivity, Personalisation, Simulation

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1 INTRODUCTION

The world of intelligent personal assistants (IPAs) [11, 16] is rapidly changing. Today's IPAs require us to initiate actions, limiting their ability to anticipate our needs and proactively help us get things done [2, 12]. Proactive Intelligent Personalised Agents (PIPAs) represent the next step in this evolution. Imagine an assistant who not only follows your commands but also proactively anticipates your needs and takes steps to complete tasks for you. By learning from your behaviour and preferences through vast amounts of user data, PIPAs have the potential to significantly improve your

daily experience and efficiency. However, this exciting potential for proactive assistance comes with a critical challenge: striking a balance between PIPAs' proactiveness and user control. Granting these powerful agents high levels of autonomy raises concerns about unintended consequences due to limitations in machine learning [9] and ultimately, user trust [22]. Imagine a PIPA booking a flight based on past travel patterns without considering a sudden change in your health, or making a significant financial decision autonomously. These scenarios highlight the potential risks associated with high PIPA autonomy.

Human-in-the-loop (HITL) control [8] emerges as a promising solution to address this challenge. HITL ensures user involvement in decision-making processes despite the potentially autonomous nature of PIPAs [2]. This research delves into the potential of HITL control as a mechanism to mitigate the risks associated with high PIPA autonomy. We focus on two key questions:

- (1) Triggering HITL: When and under what circumstances should HITL control be triggered to ensure user involvement in critical decisions?
- (2) Impact of PIPA Autonomy: How do varying levels of PIPA autonomy, controlled by HITL mechanisms, affect their efficacy in different agent-user and agent-agent interaction scenarios?

To explore these questions, we leverage a simulation-based approach validated through a travel assistance use case where PIPAs proactively assist users in selecting the optimal travel mode based on users' preferences and behaviours. This use case choice allows for accurate simulation of diverse PIPA components across various contexts while avoiding the challenges associated with user preference data collection. Publicly available national travel surveys, frequently conducted in metropolitan areas, provide this data. These surveys offer valuable insights into user travel behaviour, informing policy decisions that promote sustainable transportation choices. The use case exemplifies HITL control integration within PIPAs, balancing agent autonomy with user control for critical travel decisions.

2 RELATED WORK

Intelligent personal assistants (IPAs) [11, 16] have garnered significant research interest, but integrating proactive behaviour into these assistants and effectively addressing user control and personalisation mechanisms remains an under-explored area. This section reviews related work, highlighting the limitations in user control and personalisation within proactive IPAs. An exemplary work by Pejovic and Musolesi [14] details stages for proactive intervention generation (inferring context, predicting future context,



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and generating interventions). However, users lack control over the frequency, type, and timing of these interventions, hindering user acceptance. Additionally, the work does not consider user goals, limiting the ability of the system to act proactively on the user's behalf. Meurisch et al. [12] introduce a reference model for achieving proactivity in IPAs that incorporates user goals. This complements Pejovic and Musolesi's work [14] but lacks a comprehensive user representation. The model focuses on six general areas for assistance (physical, social, etc.) but omits representing individual user preferences within the model. Yorke-Smith et al. [21] introduce a proactive personal agent design based on the belief–desire–intention (BDI) model of agency [7]. While their design incorporates goal generation, it lacks clarity on how user control could be integrated and how it could influence the level of autonomy granted to agents.

The works from the literature we have highlighted and cited above are single-agent systems. However, to expand our understanding of proactivity in IPAs, we have also explored research on multi-agent systems. Two relevant works exploring proactivity within this domain are presented here. The first work, the Electric Elves project by Tambe et al. [19], deployed a dozen IPAs in an office setting for several months. Each employee had an IPA assigned to manage routine tasks like scheduling meetings. The project's primary focus was on team coordination among the agents. However, it also identified crucial issues such as privacy, social norms, and a concept called adjustable autonomy. Adjustable autonomy [15] is particularly relevant to our research. It allows agents to dynamically adjust their decision-making based on user input. Tambe et al. present three choices for adjustable autonomy: (i) autonomous decision, where the agent independently takes action; (ii) transfer control to the user, allowing the user to make decisions; and (iii) adjust coordination constraints, where the agent adapts its decision-making parameters, such as extending time allowances for user input during scheduling meetings. While Tambe et al. illustrate this concept with a meeting scheduling scenario, their work lacks details on several crucial aspects. Firstly, it is unclear how proactivity is integrated into the agent (agent architecture). Secondly, the mechanism for integrating user control and determining when/how to trigger it remains unexplained. Thirdly, the methods for building user models and training the agents on them (user modeling and training data) are not discussed. Finally, the sources of data used for agent training and decision-making are not mentioned. The second work, by Akbar [2], explores agent proactivity in a multi-agent meeting scheduling scenario. Similar to Tambe et al.'s research [19], it lacks details about several key areas. The composition and functionalities of the agent components (agent architecture) are not presented. Additionally, while Human-in-the-loop (HITL) implementation is claimed, it functions as a simple on/off switch transferring control to the user with a fixed delay. There is no reasoning mechanism to determine when to activate HITL or if an autonomous decision is suitable.

The limitations identified in existing research highlight a significant gap in the architecture and decision-making processes within proactive IPAs. We lack a clear understanding of how agents integrate proactivity, implement user control and personalisation mechanisms, and utilise data for user modeling and training. While user control has been explored in fields like autonomous vehicles

[8], autonomous weapons [5], and industry 4.0 [4], directly applying those methodologies to the field of IPAs proves challenging due to the vastly different fields. Furthermore, these works [4, 5, 8] do not address our core questions: (1) when and how to trigger HITL control and (2) how varying levels of agent autonomy, regulated by HITL, impact agent's efficacy in different situations. This gap presents a crucial opportunity to develop robust and user-centric proactive IPAs that can effectively leverage adjustable autonomy and user control for proactive task management.

3 TOWARDS A SIMULATION-BASED APPROACH TO DEVELOP PIPAS

This work proposes a simulation-based approach for designing and developing the Proactive Intelligent Personal Assistants (PIPAs). The approach offers several unique advantages:

Manipulation of Contextual Factors: Controlling external contextual factors such as weather and traffic is impossible in real-world experiments of travel assistance use cases. However, a simulation-based approach allows us to create controlled environments where we manipulate these contextual factors including the user-specific contextual factors such as user location, and schedule, etc to observe their impact on the agent's ability to predict user intent.

Exploration of Diverse Scenarios: Simulations enable the creation of diverse scenarios across extended time frames. Unlike real-world limitations, where user engagement beyond short periods is challenging, simulations can seamlessly test agent performance over a 24-hour cycle (t1-t24). We can define how these contextual factors (location, weather, traffic) change throughout the day, mimicking real-world dynamics. This controlled environment also allows us to introduce unforeseen events – unexpected traffic delays or sudden rain showers – to assess the agent's adaptability. Imagine a user's work meeting runs late, or a planned cycling trip encounters downpours. Simulations let us test how the agent responds by suggesting alternative travel options in real-time (adjusting travel mode based on traffic or proposing rain-friendly alternatives for cycling). Importantly, simulations allow us to create scenarios with extreme conditions (e.g., mass transit disruptions due to infrastructure issues, weather, maintenance, labor actions, or unforeseen events) before they happen in the real world. This allows us to ensure the robustness of the agent before deployment.

Exploration of Coordination and Negotiation in Multi-Agent Systems: Simulations offer a powerful tool to investigate the behaviour of complex Multi-Agent Systems (MAS) with hundreds of interacting agents. These agents engage in two key social behaviours: coordination for collaborative task execution and negotiation for joint activity scheduling. In our travel assistance use case, MAS involves agents representing users who negotiate travel times (based on calendar data) and modes of transport (based on user travel preferences). Additionally, coordination ensures users arrive on time at a central location (e.g., bus station, train station, airport). Agents would share information regarding potential delays and collaboratively adjust arrival times to maintain group cohesion. With hundreds of agents interacting, unforeseen emergent behaviours might arise. Simulations allow us to observe how the agents behave in various situations, e.g. unexpected traffic delays or changes in user preferences. This helps identify and address potential issues

that might be difficult to predict in the real world. Simulations provide a controlled environment to gather vast amounts of data on agent interactions, negotiation outcomes, and travel mode selections. Analysing this data helps refine the negotiation protocols, optimise coordination strategies, and personalise recommendations for different user groups.

Cost-Effectiveness and Privacy-Preservation: Simulations mitigate the substantial economic and ethical challenges associated with the deployment of agents in real-world contexts. One critical advantage of simulations is their ability to circumvent the privacy concerns associated with the use of survey data on user preferences and the continuous monitoring of users' contexts, such as location and calendar events. Prior research, including the Electric Elves project [19], has highlighted the privacy risks that can emerge with the deployment of agents in real-world scenarios. Furthermore, studies involving real users are fraught with challenges: they are typically costly, time-consuming, and plagued by difficulties in participant recruitment [13]. By contrast, simulations eliminate the need for expensive field deployments by utilising synthetic user profiles, thereby addressing the logistical challenges and reducing the costs associated with recruiting and retaining participants for longitudinal studies [3].

The following subsections illustrate how simulation offers the aforementioned benefits in the context of developing PIPAs for travel assistance use cases.

3.1 Creating Synthetic Users for PIPAs Development

In our travel assistance use case, we leverage user travel survey data collected by the London transportation authority (TfL) and transformed into a dataset by Tim et al [10]. This rich dataset contains travel details for thousands of users, including user characteristics and trip details, allowing us to understand user travel behaviour based on various factors. We create synthetic users by analysing user behaviour patterns within the data. First, we cluster users based on similarities in their travel choices. Then, we utilise these clusters to fabricate synthetic users that represent the real-world distribution of travel behaviour in the dataset. This simulation approach allows us to generate a vast pool of synthetic users with diverse travel preferences and behaviours, mirroring real passengers in a metropolitan city. Each synthetic user profile becomes the basis for a user model, enabling the PIPA to adapt its behaviour based on predicted user preferences.

3.2 Enabling Proactive Behaviour in PIPAs

Proactive behaviour is essential for PIPAs to effectively anticipate user needs. Integrating this functionality remains a challenge, but advancements in context-aware computing offer promising solutions [14]. Context-aware systems leverage contextual information (e.g., user location, activity) to deliver relevant information and services [20]. This concept of context, broadly defined as "any information characterizing an entity's situation" [1], empowers PIPAs to proactively identify task needs within a specific context [17, 18]. Crucially, user context can change rapidly, influencing task preferences. For instance, a user's desire to complete a task or their preferred approach might shift due to a context change. Therefore, continuous monitoring of user context is vital for effective intent

prediction which is key to enabling proactive behaviour. Prior research emphasises the importance of context in anticipating user intent [17, 18]. Understanding user intent is crucial for goal-based decision-making and proactive support [2, 12]. Therefore, our approach focuses on modeling user context and leveraging it to predict real-time user intent, enabling the activation of proactive behaviour in PIPAs.

Context collection and modeling involve gathering users' contextual information through sensors and APIs. This includes real-time data such as location, calendar schedules, and environmental factors (weather, traffic). While real sensors and APIs offer valuable data, they require significant effort for processing and context modeling. To overcome this, we propose utilising simulated APIs. These allow for cost-effective testing under various contextual conditions without the complexity of real-world data processing. Simulated APIs can be designed to provide context in a pre-defined format, eliminating the need for extensive data processing and model development. The agent's intent predictor component then utilises this modeled user context to predict users' contemporary intent by extrapolating preferences and behaviour from the user models built from synthetic user profiles.

3.3 Integrating Human-in-the-loop Control in PIPAs

User control is central to our vision for PIPAs. However, striking a balance between agent autonomy and user control is a complex challenge. Uncertain user intent makes autonomous decisions risky, while frequent user interruptions can be disruptive. Our research explores strategies for navigating this trade-off. We propose a human-in-the-loop (HITL) approach [8] to ensure users remain involved in decision-making despite the potentially autonomous nature of PIPAs. Effective HITL integration hinges on determining when and how to trigger user involvement. However, this is not a simple on/off switch, but rather a nuanced process considering several factors. Three key factors influence the need for HITL control:

3.3.1 User Preferences for Agent Autonomy. User preferences for agent autonomy vary by situation. For example, a user running late for a meeting might grant full control to book a taxi, even if slightly expensive, to ensure on-time arrival. However, this preference might differ for leisure travel, where cost is a bigger concern. While traditional user surveys can gather preference data, this approach is not feasible for our synthetic users whose behaviour is based on real user data. There is no existing data on their preferred level of agent autonomy in different situations. One approach is to generate synthetic user preference data through scripts that consider input parameters such as autonomy levels users might grant agents in various scenarios. However, validating the quality of this synthetic data remains a challenge due to the lack of a baseline for comparison. Overall, this functionality involves considering user preferences in determining whether to autonomously book the chosen mode or transfer control to the user. User preferences can inform the creation of decision rules. For example, (i) if the trip is work-related and risks a late arrival, the agent can autonomously book a faster mode, and (ii) if the estimated cost falls below a user-defined threshold (e.g., 30 euros), the agent can make an autonomous decision.

3.3.2 Cost Factors. It is crucial for PIPAs to carefully assess various cost factors associated with autonomous decision-making and the transfer of control to the user through HITL. Firstly, the PIPA should evaluate the *cost of autonomous decisions*, which directly influences the necessity for HITL intervention. This encompasses the immediate expenses tied to the action taken, such as booking a taxi. Notably, decisions with higher costs, like arranging long-distance travel, often warrant user involvement compared to simpler tasks like local transportation bookings. Costs here can be measured in both time and money, with user preferences varying between prioritising speed and affordability. The agent must account for these preferences when considering decision options. Secondly, the PIPA should assess the *cost of erroneous actions*, encompassing the repercussions of mistakes made by the agent, such as selecting the wrong mode of travel. Thirdly, it is essential to evaluate the *cost of user handoff*, which refers to the disruption caused by involving the user in decision-making. Considering the user's ongoing tasks and potential interruptions is crucial to minimise the cost of hand-offs. Lastly, the PIPA should consider the *cost of miscoordination*. In multi-agent systems, where multiple agents operate on behalf of their users, transferring control to the user and then waiting for user input can lead to significant inefficiencies. Agents become interdependent when their actions rely on each other's decisions. Delays caused by one agent waiting for user input can cascade and impact other agents waiting for a response. If one agent is stuck waiting for a user, it can block other agents from making progress, creating a domino effect of delays throughout the system. These delays ultimately reduce the overall throughput and efficiency of the multi-agent system.

Analysing miscoordination costs directly addresses our second research question: how varying levels of PIPA autonomy, controlled by HITL mechanisms, affect agent efficacy in different interaction scenarios (both agent-user and agent-agent). While it is anticipated that HITL-controlled user intervention may lead to miscoordination in multi-agent interactions, empirical experimentation is needed to confirm this hypothesis. Furthermore, in a single-agent system, if control is transferred and the user does not respond within a set timeframe (e.g., 5 minutes), the agent must make an appropriate autonomous decision. However, determining this optimal timeframe is challenging. Should it depend on the urgency of the decision or individual user behaviour? Furthermore, transferring control to avoid decision-making costs might lead to delayed decisions with their costs. For instance, if the user responds within the timeframe but the travel window has closed, the agent might have to rebook at a higher price. How can we effectively manage these competing factors i.e. costs of waiting and autonomy? Effectively managing these competing factors requires encoding initial knowledge about costs and uncertainties into the agent. Frameworks such as Markov Decision Processes [6] can model decision-making for various scenarios, from booking to canceling travel arrangements.

3.3.3 Uncertainty in User Intent. A critical factor for HITL activation is uncertainty about user intent. In situations where the agent is unsure of the user's preference between two travel modes, its confidence in the optimal choice will be low. This low confidence triggers HITL. We propose using confidence scores to quantify uncertainty and determine HITL activation. These scores can be

calculated based on the probability difference between the top two travel mode recommendations from the prediction model. If the probability difference between the top two modes is significant (above a threshold), the agent is considered confident, and HITL is not triggered. A low confidence score (small probability difference) indicates high uncertainty. In such cases, HITL is triggered, involving the user in the decision-making process. This approach extends beyond two options. When considering more than two travel modes, alternative methods like calculating variance can be explored to assess the agent's confidence level.

Simulation can be a valuable tool in this context. We can gain valuable insights into agent's behaviour by simulating a large user pool with diverse preferences for agent control and travel mode choices. This allows us to design a HITL control mechanism that adapts to individual user needs and preferences.

4 DISCUSSION AND CONCLUSION

This research uses a simulation-based approach to explore the integration of Human-in-the-loop (HITL) control within Proactive Intelligent Personalised Agents (PIPAs). It leverages the power of simulations to control and manipulate contextual factors, explore a vast range of scenarios efficiently, examine multi-agent interactions, ensure cost-effectiveness, and preserve privacy. The two primary questions addressed in this research are: (1) what are the optimal conditions under which HITL control should be triggered, and (2) how different levels of agent autonomy, governed by HITL control, impact the agents' efficacy across various scenarios. For the first question, we propose that HITL activation should be guided by three main considerations: (i) user preferences regarding agent autonomy (accommodated by deriving decision rules from user surveys), (ii) cost implications including the costs associated with autonomous decisions, errors, user interventions, and coordination failures (modeled using frameworks such as Markov Decision Processes (MDPs) [6]), and (iii) the level of uncertainty in agent's prediction of user intent (quantified through confidence scores calculated from the probability differences between the highest probability travel options; if the difference exceeds a specified threshold, indicating high confidence, HITL is not triggered, but a low score signifies uncertainty and triggers user involvement in decision-making). The second question relates to the costs identified in the first question, particularly the costs associated with miscoordination. In scenarios involving multiple agents, where different users exert varying levels of control, coordination issues are expected due to agents potentially needing to pause and wait for user responses, thereby impacting collaborative and negotiation tasks. Similarly, in single-agent systems, delays in user response could lead to missed opportunities, such as failing to book a transport mode or incurring higher costs due to delayed booking. These factors could significantly affect the efficacy of the agents. However, experimental evidence is required to substantiate this hypothesis.

This research acknowledges limitations associated with creating synthetic users based on publicly available travel survey data. While this data adequately serves the development of PIPAs in London, benefitting from comprehensive surveys conducted by the London transport authority [10] encompassing a diverse array of influential factors on travel mode selection, its applicability to other

cities, such as Dublin or Paris, may be limited. This stems from the absence of key factors in survey data obtained from the Ireland transport authority¹ or the French transport authority², thereby constraining its utility for accurately modeling user choices in those locations. In such cases, synthetic data generation is necessary. However, this approach presents challenges in ensuring validity. A simple rule-based approach with predefined conditions for travel mode selection may not adequately reflect real-world user behaviour. Furthermore, publicly available user travel datasets typically do not include preferences regarding agent autonomy, making it difficult to align synthetic data for both travel mode choice and autonomy preferences. Additional challenges include the absence of contextual information (e.g., location, calendars, weather) in user data from travel surveys. These factors significantly influence travel decisions. While including user context factors in real data is problematic due to privacy concerns, purely synthetic context data also presents evaluation challenges. This highlights a limitation in achieving perfect simulation fidelity. While simulations have numerous advantages, they may struggle to fully capture the intricacies of human decision-making, which can be unpredictable and evolve over time. Consequently, a PIPA developed solely in a simulation environment might not perfectly translate to the real world. However, the challenge of capturing human decision-making complexities might not be unique to simulations. It is possible that even in real-world operational contexts, perfectly predicting unseen user behaviour remains a challenge. After all, real users can be unpredictable in their decision-making, and their preferences and behaviours can evolve over time. While simulations may not fully capture these intricacies, they importantly offer the capability to introduce randomness in user behaviour and decision-making. This allows us to test the agent's robustness in a controlled environment.

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¹<https://data.cso.ie/>

²<https://mobiliscope.cnrs.fr/en>